# The Selected Higher-Star Defect of a Design Incidence Lattice

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**Abstract**

For a nonempty $t$-$(v, k, \lambda_t)$ design with $2 \le t \le k < v$, let $A$ be its point-by-block incidence matrix. We study the finite quotient $\Delta(D) = H_k / \operatorname{im}_{\mathbb{Z}}(A)$, where $H_k$ consists of the integer point vectors whose coordinate sum is divisible by $k$. A pointed presentation embeds the dual quotient isotropically in a bilinear discriminant group determined by the design parameters. This gives an odd-prime bound that improves square divisibility by one generator on an infinite family of $S(2, 5, v)$, together with a rank-three reduction at exceptional primes. At the prime two, a general bilinear lemma shows that isotropy gives no further bound on the order. We also prove that a Pasch switch preserves the odd-primary quotient and changes its order by at most a factor of two. Finally, published exact-rank bounds and an explicit asymptotic comparison realize every fixed elementary binary or ternary group at all sufficiently large orders in the corresponding Steiner triple-system families. Classical incidence identities, Smith forms and enumeration results provide the inputs; four finite examples are reproduced by a companion verifier.

## 1 Introduction

The incidence matrix of a block design records which points belong to each block. Its integral column lattice describes the integer point vectors that can be assembled from blocks. Every such vector has coordinate sum divisible by the block size, but this necessary condition need not be sufficient. The quotient measuring this failure is the object of this paper.

Let $D = (P, \mathcal{B})$ be a nonempty, possibly non-simple $t$-$(v, k, \lambda_t)$ design with $2 \le t \le k < v$. Write $A$ for its point-by-block incidence matrix and set

$$L = \operatorname{im}_{\mathbb{Z}}(A), \qquad H_k = \Big\{ z \in \mathbb{Z}^P : \textstyle\sum_{x \in P} z_x \equiv 0 \pmod{k} \Big\}.$$

Since every column has weight $k$, one has $L \le H_k$.

**Definition 1.1.** The *selected higher-star defect* of $D$ is the quotient $\Delta(D) = H_k / L$.

The word *selected* refers to the actual blocks of $D$, rather than the complete collection of $k$-subsets. Theorem 2.1 shows that the defect is finite. It is the kernel of the coordinate-sum map $\operatorname{coker}(A) \to \mathbb{Z}/k$, so it removes the cyclic factor forced by the common block size. We study its bounds, its dependence on the chosen design and its behavior under local changes of blocks.

### 1.1 Bounds from the design parameters

Classical star identities give an annihilator obtained from the successive incidence numbers. After this elementary reduction in Section 2, we replace $A$ by a pointed matrix presenting the defect itself. Its Gram matrix depends only on the ordinary 2-design parameters. The dual defect then sits as an isotropic subgroup of the associated finite bilinear group (Proposition 3.1).

This presentation gives two useful local calculations. At odd primes of valuation one, an explicit determinant class yields a Witt bound that is strictly stronger than square divisibility on an infinite family of $S(2, 5, v)$ (Theorem 4.2). When the usual zero-sum decomposition is unavailable, a rank-three matrix contains the remaining local information (Theorem 4.3). For binary subgroup orders, the answer is simpler: a general bilinear lemma shows that isotropy alone gives no improvement on the square-divisibility bound (Corollary 4.5).

### 1.2 Variation, local changes and realization

The parameters do not determine the defect. Two Steiner triple systems on 15 points have the same parameter Gram matrix but different selected quotients (Theorem 5.2). When a symmetric design has order of $p$-adic valuation two, however, one modular rank determines the local group (Theorem 5.1). Section 5 treats these two phenomena.

A Pasch switch replaces four triples by four others on the same six points. Theorem 6.1 shows that it preserves every odd-primary part of the defect and changes the total order by a factor in $\{1/2, 1, 2\}$. The proof follows the two local column lattices after the unchanged blocks have been adjoined.

The realization result has an infinite scope: every fixed elementary binary or ternary group occurs for all sufficiently large orders in the respective classical Steiner triple-system families (Theorem B.1). Appendix B proves the required asymptotic comparison of published exact-rank counts. The geometric applications and worked examples show both vanishing and nonzero defects.

### 1.3 Relation to earlier work

The higher-star identity comes from Wilson's incidence theory [1, 2]. The universal additive group and the $t = 2$ point-difference identity are treated by Caggegi–Falcone–Pavone [5]. Later developments concern additive embeddings of designs [6, 7]; the point-injectivity questions in that theory are outside the scope of this paper.

The complete-subset incidence matrix has a classical diagonal form [3], and integral design modules are studied in [23, 24, 20]. Here the column lattice belongs to a particular design. We use the standard theory of finite bilinear forms and lattice saturation [10, 11]; applying Witt theory to designs also has classical precedents [4, Sections 5–6].

The deductions developed below are the explicit pointed Gram calculation, its prime-local consequences, the Pasch quotient law and the fixed-defect asymptotic comparison. The star bound and binary order lemma are elementary supporting facts. Geometric Smith forms, finite rank spectra and enumeration theorems are imported where they are used; Sin's survey [8] provides broader context for incidence Smith forms. These results concern the image lattice and bilinear isotropy. They neither classify relation kernels nor settle general realization or quadratic-refinement questions.

### 1.4 Notation

For $1 \le j \le t$, let $\lambda_j$ be the number of blocks containing a fixed $j$-set. We use

$$r = \lambda_1, \qquad \lambda = \lambda_2, \qquad \alpha = r - \lambda, \qquad n = v - 1, \qquad b = |\mathcal{B}|.$$

For a finite abelian group $F$, write $F_p$ for its $p$-primary part, $\ell_p(F) = \log_p |F_p|$ and $\mu(F)$ for its minimum number of generators. The symbols $\mathbb{Z}_p$ and $\mathbb{F}_p$ denote the $p$-adic integers and the field of $p$ elements. Unless stated otherwise, a discriminant pairing is symmetric and bilinear with values in $\mathbb{Q}/\mathbb{Z}$.

## 2 The higher-star bound

The successive incidence numbers give the integer

$$d = \gcd\{\lambda_j - \lambda_{j+1} : 1 \le j < t\}.$$

It controls the exponent of the selected quotient.

**Theorem 2.1.** *With the notation above,*

$$d\,H_k \;\le\; L \;\le\; H_k.$$

*Consequently $\Delta(D) = H_k/L$ is finite, is killed by $d$, is generated by at most $v-1$ elements, and has order dividing $d^{\,v-1}$. There is a canonical exact sequence*

$$0 \to \Delta(D) \to \operatorname{coker}(A) \to \mathbb{Z}/k \to 0,$$

*and, for the retained-apex matrix $A^+ = \left[\begin{smallmatrix} \mathbf{1}^T \\ A \end{smallmatrix}\right]$, a canonical equality $\Delta(D) = \operatorname{Tor}\operatorname{coker}(A^+)$; the splitting $\operatorname{coker}(A^+) \cong \mathbb{Z} \oplus \Delta(D)$ holds noncanonically.*

*Proof.* Column weight $k$ gives $L \le H_k$.

For the lower bound, let $u_R$ be the sum of all block columns containing a fixed $j$-set $R$; coordinate counting gives $u_R = \lambda_{j+1}\mathbf{1}_P + (\lambda_j - \lambda_{j+1})\mathbf{1}_R$. Hence, for distinct points $x, y$ and an $(j-1)$-set $S$ disjoint from them,

$$u_{S\cup\{x\}} - u_{S\cup\{y\}} = (\lambda_j - \lambda_{j+1})(e_x - e_y) \in L,$$

and Bézout over $1 \le j < t$ yields $d(e_x - e_y) \in L$ for all $x, y$. Fix a block $B$ and a point $o \in B$; since $d\,\mathbf{1}_B \in L$,

$$d\,\mathbf{1}_B - \sum_{x\in B\setminus\{o\}} d(e_x - e_o) = dk\,e_o \in L.$$

The vectors $\{e_x - e_o : x \ne o\}$ together with $k\,e_o$ generate $H_k$, and both $d(e_x - e_o)$ and $dk\,e_o$ lie in $L$, so $dH_k \le L$. Thus $\Delta(D)$ is finite and killed by $d$.

In $H_k/L$ the block relation $\mathbf{1}_B = k\,e_o + \sum_{x\in B\setminus\{o\}}(e_x - e_o)$ vanishes, expressing the class of $k\,e_o$ through the $v-1$ point-difference classes; hence $\Delta(D)$ is a quotient of $(\mathbb{Z}/d)^{\,v-1}$, so it has at most $v-1$ generators and $|\Delta(D)|$ divides $d^{\,v-1}$.

The coordinate-sum map $\operatorname{coker}(A) \to \mathbb{Z}/k$, $[z] \mapsto \sum z \bmod k$, has kernel $H_k/L = \Delta(D)$ and is onto because $[e_x] \mapsto 1$ generates $\mathbb{Z}/k$.

Finally the primitive map $\mathbb{Z} \oplus \mathbb{Z}^P \to \mathbb{Z}$, $(a, z) \mapsto ka - \sum z$, has kernel identified with $H_k$ by $z \mapsto (\sum z/k, z)$. The columns of $A^+$ belong to that kernel. Quotienting by them gives $0 \to \Delta(D) \to \operatorname{coker}(A^+) \to \mathbb{Z} \to 0$, which splits noncanonically and identifies $\Delta(D)$ canonically with the torsion subgroup. □

The star mechanism (the identity producing the coefficients $\lambda_r - \lambda_{r+1}$) is a specialization of Wilson's higher-incidence identity [1]; see [2, Section 2.2, equation (2.1)] with $e = 1$ and $f = j$. The universal group presented by $A$, and the $t = 2$ point-difference identity, are in [5, Section 2]. The gcd over star levels and the retained-row passage above are elementary deductions.

**Corollary 2.2.** *If $d = 1$ then $L = H_k$ and $\Delta(D) = 0$, i.e. $\operatorname{coker}(A) \cong \mathbb{Z}/k$ and the full incidence Smith form is $1^{\,v-1}k$.*

## 3 A pointed presentation and its Gram matrix

Fix $x_0 \in P$. Deleting the row $x_0$ from $A$ and appending the all-ones block row gives a full-row-rank $v \times b$ matrix $C_{x_0}$; the basis $\{e_x - e_{x_0}\} \cup \{ke_{x_0}\}$ of $H_k$ turns each incidence column into its coordinates away from $x_0$ followed by a final 1.

**Proposition 3.1.** *$\Delta(D) \cong \operatorname{coker}(C_{x_0})$, and different choices of $x_0$ differ by unimodular row operations.*

*The Gram matrix $G = C_{x_0}C_{x_0}^T$ depends only on the ordinary 2-design parameters,*

$$G = \begin{pmatrix} \alpha I_n + \lambda J_n & r\,\mathbf{1}_n \\ r\,\mathbf{1}_n^T & b \end{pmatrix}, \qquad \det G = \alpha^{\,n}\, r/k,$$

*so* $|\Delta(D)|^2$ *divides* $\alpha^n r/k$*, and for every prime* $p$*,* $2\,\ell_p(\Delta) \le n\,v_p(\alpha) + v_p(r) - v_p(k)$*.*

*Writing* $R = \mathrm{im}(C_{x_0}^T)$ *and* $\bar{R} = (R \otimes \mathbb{Q}) \cap \mathbb{Z}^b$*, the group* $\bar{R}/R$ *is canonically Pontryagin dual to* $\Delta(D)$ *and embeds in* $\mathrm{coker}(G) \cong R^{\#}/R$ *as a totally isotropic subgroup for the bilinear discriminant form* $\beta_G(z,w) = z^T G^{-1} w \bmod \mathbb{Z}$*. (No quadratic refinement is asserted without parity hypotheses.)*

*Proof.* The pointed basis identifies $H_k/L$ with $\mathrm{coker}(C_{x_0})$; the Gram computation is the standard 2-design identity, and $\det G = \alpha^n r/k$ follows from $(rk-\lambda)b - nr^2 = \alpha r/k$.

Since $C_{x_0}$ has full row rank $v$, its finite cokernel has order $|\Delta|$ equal to the greatest common divisor of the $v \times v$ minors of $C_{x_0}$ (the top determinantal divisor $d_v$). By the Cauchy–Binet formula, $\det G = \det(C_{x_0} C_{x_0}^T)$ equals the sum of the squares of those maximal minors; each minor is divisible by $|\Delta|$, so every summand is divisible by $|\Delta|^2$, whence $|\Delta|^2 \mid \det G$. Taking $p$-adic valuations gives the primewise bound $2\ell_p(\Delta) \le v_p(\det G) = n\,v_p(\alpha) + v_p(r) - v_p(k)$.

For completeness, the duality pairing is explicitly

$$(\bar{R}/R) \times (\mathbb{Z}^v / C_{x_0}\mathbb{Z}^b) \longrightarrow \mathbb{Q}/\mathbb{Z}, \qquad (C_{x_0}^T q + R,\, z + C_{x_0}\mathbb{Z}^b) \longmapsto q^T z \bmod \mathbb{Z}.$$

The vector $q \in \mathbb{Q}^v$ is unique, since $C_{x_0}$ has full row rank. Changing either representative changes the value by an integer. Smith normal form shows that this pairing is perfect, hence identifies $\bar{R}/R$ with $\Delta^\vee$.

Finally $\bar{R} \subseteq \mathbb{Z}^b$ has integral dot products, so $\bar{R} \subseteq R^{\#}$ and the induced discriminant pairing vanishes on $\bar{R}/R$. This is the bilinear saturation construction underlying the overlattice framework [11]; no even-lattice assumption is used. □

Thus the abstract annihilation of Theorem 2.1 becomes a computable structural invariant: a finite bilinear form determined by the parameters, with the dual defect sitting inside it isotropically.

# 4 The discriminant form prime by prime

Throughout this section $E_p = \mathrm{coker}(G)_p$ with the pairing $\beta_G$, and $\ell_p$ denotes length. The maximal totally isotropic subgroup of $E_p$ bounds $\Delta_p$ from above; the square-order bound of Proposition 3.1 is the statement that this maximum is at most $|E_p|^{1/2}$.

## 4.1 Good primes

**Proposition 4.1** (good-prime decomposition)**.** *For every prime* $p \nmid n$*, the pointed lattice splits* $\beta_G$*-orthogonally as*

$$E_p \;\cong\; (W_p/\alpha W_p,\ (x,y) \mapsto x\cdot y/\alpha) \;\perp\; (\operatorname{coker} K_p,\ K_p^{-1}), \qquad K_p = \begin{pmatrix} n(rk-\lambda) & nr \\ nr & b \end{pmatrix},$$

*where* $W_p = \{x \in \mathbb{Z}_p^n : \mathbf{1}^T x = 0\}$*; here* $\det K_p = n\alpha r/k$*.*

*Proof.* Since $p \nmid n$, the vector $\mathbf{1}_n$ has unit norm $n$, so $\mathbb{Z}_p^n = W_p \perp \mathbb{Z}_p \mathbf{1}_n$; adjoining the apex coordinate $\mathbb{Z}_p e_*$ gives the orthogonal splitting $\mathbb{Z}_p^v = W_p \perp T_p$ with $T_p = \mathbb{Z}_p \mathbf{1}_n \oplus \mathbb{Z}_p e_*$. Restricting $G$ from Proposition 3.1: on $W_p$ (coordinate sum zero) it is $\alpha$ times the dot product, which is unimodular there; on $T_p$ it is the block $K_p$, whose determinant $n\alpha r/k$ follows from $(rk-\lambda)b - nr^2 = \alpha r/k$. Passing to cokernels and inverse-Gram pairings in this orthogonal basis gives the stated sum; $\ell_p$ of each summand is read from the entry valuations of $K_p$. □

## 4.2 The odd-prime Witt bound

The determinant class distinguishes the two possible Witt indices in even dimension. The following calculation makes that class explicit.

**Theorem 4.2** (odd valuation-one Witt obstruction)**.** *Let $p$ be odd with $p \nmid k\lambda$ and $v_p(\alpha) = 1$, and suppose $n = v - 1$ is even, $n = 2m$. Then $E_p \cong \mathbb{F}_p^n$ carries a nondegenerate symmetric form, its dual selected defect is totally isotropic, and its minimal number of generators $\mu(\Delta_p)$ satisfies*

$$\mu(\Delta_p) \le m \quad \text{if } (-1)^m k \text{ is a square mod } p, \qquad \mu(\Delta_p) \le m-1 \quad \text{otherwise.}$$

*The second line is one generator stronger than $|\Delta_p|^2 \mid \det G$. For $v = 5 + 60t$ with $t \ge 1$ and $3 \nmid t$, Hanani's theorem [14] supplies an $S(2, 5, v)$; then $n = 4 + 60t$ is even, at $p = 3$ one has $m = 2 + 30t$, $(-1)^m k = 5$ a nonsquare, so $\Delta_3 \cong (\mathbb{Z}/3)^\mu$ with $\mu \le 1 + 30t$, whereas the coarse envelope permits $2 + 30t$. This is an infinite strict obstruction.*

*Proof.* The hypotheses give $r \equiv \lambda \pmod p$. The identities $r(k-1) = \lambda(v-1)$ and $bk = vr$ then give $v \equiv k$ and $b \equiv \lambda \pmod p$, respectively. Thus $G \equiv \lambda J_v \pmod p$ has rank one, and $v_p(\det G) = n$ forces every nonunit invariant factor to have valuation one, so $E_p \cong \mathbb{F}_p^n$. Identify $(1/p)\mathbb{Z}/\mathbb{Z}$ with $\mathbb{F}_p$ by $a/p \mapsto a \bmod p$. Direct inversion gives

$$G^{-1} = \frac{1}{\alpha}\begin{pmatrix} I_n + J_n & -k\mathbf{1}_n \\ -k\mathbf{1}_n^T & k(kr-\lambda)/r \end{pmatrix}.$$

The first $n$ coordinate classes form a basis of $E_p$: the image of $G$ modulo $p$ is the constant line in $\mathbb{F}_p^v$. In this basis the induced finite-field form has matrix $u^{-1}(I_n + J_n)$, where $u = \alpha/p$. Its determinant is $v/u^n \equiv k/u^n \pmod p$, of square class $k$ for even $n$.

By the classical determinant criterion [4, Theorem 5.2] (see also [13]), a $2m$-dimensional such space has Witt index $m$ iff $(-1)^m$ times its determinant is a square, else $m-1$; the dual defect is a totally isotropic subspace, so its dimension is at most the Witt index. The $S(2, 5, v)$ evaluation is direct. □

## 4.3 Exceptional primes

For $p \mid n$ the root/line split fails; a different primitive orthogonal splitting survives.

**Theorem 4.3** (exceptional-prime reduction)**.** *For every prime $p \mid n$ the pointed lattice is $\mathbb{Z}_p$-congruent to $\alpha D_{n-2} \perp C_p$, where $D_{n-2} = I + J$ has rank $n-2$ and determinant $n-1$ (a $p$-adic unit, so $D_{n-2}$ is unimodular over $\mathbb{Z}_p$), and*

$$C_p = \begin{pmatrix} \alpha n/(n-1) & -\alpha/(n-1) & 0 \\ -\alpha/(n-1) & \lambda + \alpha/(n-1) & r \\ 0 & r & b \end{pmatrix}, \qquad \det C_p = \frac{\alpha^2 r}{k(n-1)}.$$

*Hence all but at most three local invariant factors have valuation $v_p(\alpha)$, and the remaining three are read from the determinantal ideals of $C_p$. At $p = 2$ the root block $\alpha D_{n-2}$ is type II; if $v_2(\alpha) = 1$ and $k\lambda$ is odd, the whole discriminant space is symplectic of dimension $n$, so bilinear isotropy meets—but does not beat—the square-order bound.*

*Proof.* Put $t = n - 1$, a $p$-adic unit. The sublattice $U = \{x \in \mathbb{Z}_p^n : x_n = 0,\ \sum_i x_i = 0\}$ has dot-product Gram $D_{n-2} = I + J$ ($\det = n-1$, unimodular over $\mathbb{Z}_p$) and is $G$-orthogonal to $s = (\mathbf{1}_n - e_n)/(n-1)$, $a = e_n - s$, and the apex $e_*$. A direct computation gives $\langle a, a\rangle_G = \alpha n/(n-1)$, $\langle a, s\rangle_G = -\alpha/(n-1)$, $\langle s, s\rangle_G = \lambda + \alpha/(n-1)$, $\langle a, e_*\rangle_G = 0$, and $\langle s, e_*\rangle_G = r$, so in the local basis $(a, s, e_*)$ the residual Gram is exactly $C_p$; the determinant identity $\alpha^{n-2} \det D_{n-2} \det C_p = \alpha^n r/k = \det G$ together with adjugation gives $\det C_p$ and $C_p^{-1}$, and the three residual exponents

are the successive determinantal-ideal valuations. This is a unimodular local change of basis, not the invalid zero-sum-root/constant-line split (whose index is divisible by $p$).

At $p = 2$, $D_{n-2} = I + J$ has even diagonal and odd determinant, hence gives a type-II $2^{v_2(\alpha)}$-modular constituent (standard $\mathbb{Z}_2$ Jordan splitting [12]); in the valuation-one unit case $2G^{-1} \bmod 2$ is alternating, so the discriminant space is symplectic and its maximal isotropic dimension equals the square-order bound. □

### 4.4 The binary order bound

For subgroup orders the binary case has a simpler general answer, independent of the design. Write maxiso($E$) for the maximum order of a totally isotropic subgroup of a finite bilinear group $E$. A nonsingular group is *metabolic* if it has an isotropic subgroup of order $|E|^{1/2}$; this does not require that subgroup to be a direct summand.

**Lemma 4.4** (binary bilinear order bound)**.** *Let $E$ be a finite 2-group with a nonsingular symmetric bilinear pairing $\beta : E \times E \to \mathbb{Q}/\mathbb{Z}$, and put $\ell = \log_2 |E|$. Then*

$$\operatorname{maxiso}(E) = 2^{\lfloor \ell/2 \rfloor}.$$

*In particular every such group of square order is metabolic.*

*Proof.* For every isotropic subgroup $S$, nonsingularity gives $|S|\,|S^\perp| = |E|$ and $S \le S^\perp$, proving the upper bound.

If $|E| \ge 4$, there is an isotropic involution. Indeed, if $\dim_{\mathbb{F}_2} E[2] \ge 2$, the map $x \mapsto \beta(x, x)$ from $E[2]$ to $(1/2)\mathbb{Z}/\mathbb{Z}$ is linear, since $2\beta(x, y) = 0$, so it has a nonzero kernel. Otherwise $E$ is cyclic of order $2^s$, $s \ge 2$; for a generator $g$ with $\beta(g, g) = a/2^s$, its involution has self-pairing $\beta(2^{s-1}g, 2^{s-1}g) = a2^{s-2} = 0$ in $\mathbb{Q}/\mathbb{Z}$.

Let $H$ be the isotropic subgroup of order two so obtained. The induced pairing on $H^\perp/H$ is nonsingular and this quotient has order $|E|/4$. Induction, starting at orders one and two, gives an isotropic subgroup there of order $2^{\lfloor (\ell-2)/2 \rfloor}$. Its inverse image in $H^\perp$ is isotropic of order $2^{\lfloor \ell/2 \rfloor}$. □

**Corollary 4.5** (binary order consequence)**.** *For every design under consideration,*

$$\operatorname{maxiso}(E_2) = 2^{\lfloor (nv_2(\alpha) + v_2(r) - v_2(k))/2 \rfloor}.$$

*Thus bilinear isotropy gives no smaller bound on the* order *of $\Delta_2$ than the rounded square-divisibility bound. In particular, if $2 \mid n$ and $A = v_2(\alpha) \ge 1$, the decomposition of Theorem 4.3 satisfies*

$$\operatorname{maxiso}(E_2) = 2^{A(n-2)/2} \operatorname{maxiso}(\operatorname{coker} C_2).$$

*Proof.* Apply Lemma 4.4 to $E_2$ and use $\ell_2(E_2) = v_2(\det G)$. In the exceptional decomposition the root constituent has even length $A(n-2)$, so the formula for its orthogonal sum with the residual group gives the second equality. □

The rank-three reduction remains useful for invariant factors and the pairing, but is not needed for this order conclusion. Lemma 4.4 is an elementary fact about finite bilinear groups, proved here for clarity; it is not a design-specific obstruction. It does not classify isotropically embeddable subgroup types or quadratic refinements.

Metabolicity also must not be confused with hyperbolicity: the plane $V(4) = \frac{1}{4}\left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ on $(\mathbb{Z}/4)^2$ has the Lagrangian $2(\mathbb{Z}/4)^2$, but has no isotropic element of order four and hence no free Lagrangian.

# 5 What the parameters determine

The Gram matrix gives bounds on the defect, but it does not usually identify the group. This section contrasts a case determined by one modular rank with two designs whose identical parameters conceal different defects.

## 5.1 Symmetric designs of valuation two

**Theorem 5.1** (symmetric valuation two)**.** *Let $D$ be symmetric of order $w = k - \lambda$ (distinct from the global $n = v - 1$) and $\rho_p = \operatorname{rank}_{\mathbb{F}_p}(C_{x_0})$. If $v_p(w) = 2$ then*

$$\Delta(D)_p \;\cong\; (\mathbb{Z}/p)^{\,v+1-2\rho_p} \;\oplus\; (\mathbb{Z}/p^2)^{\,\rho_p-1},$$

*so the intrinsic local exponent is $p^2$ exactly when $\rho_p \geq 2$. For the calibration $\mathrm{PG}(2, p^2)$ this recovers the Chandler–Sin–Xiang value [9]; the exact $\mathrm{PG}(2, 4)$ pointed form is $1^{10}2^2 4^9$.*

*Proof.* For a symmetric design $|\Delta|^2 = (k-\lambda)^{v-1} = w^{v-1}$, so $v_p(|\Delta_p|) = (v-1)v_p(w)/2 = v - 1$; as $v_p(w) = 2$, $\Delta_p$ has exponent dividing $p^2$, say $\Delta_p \cong (\mathbb{Z}/p)^a \oplus (\mathbb{Z}/p^2)^b$. The mod-$p$ reduction of $\operatorname{coker}(C_{x_0})$ has dimension $v - \rho_p$, so $a + b = v - \rho_p$, while $v_p(|\Delta_p|) = a + 2b = v - 1$. Solving gives $b = \rho_p - 1$ and $a = v + 1 - 2\rho_p$. The $\mathrm{PG}(2, p^2)$ multiplicities are the Chandler–Sin–Xiang invariant factors [9]. □

## 5.2 Two triple systems with the same parameters

**Theorem 5.2** (same-parameter separation)**.** *The point/line design of $\mathrm{PG}(3, 2)$ and the Bose averaging system on $\mathbb{Z}_5 \times \mathbb{Z}_3$ are both simple 2-$(15, 3, 1)$ designs with the same parameter Gram matrix, yet*

$$\Delta(\mathrm{PG}(3,2)) \cong (\mathbb{Z}/2)^4, \qquad \Delta(\mathit{Bose}) = 0,$$

*with full incidence Smith forms $1^{11}2^3 6$ and $1^{14}3$. Hence ordinary parameters, and even the parameter Gram envelope, do not determine $\Delta$.*

*Proof.* Both constructions produce 35 triples covering the 105 pairs once. As $6\Delta = 0$ with 6 squarefree, the modular ranks $\operatorname{rank}_{\mathbb{F}_2}(C_{x_0})$ and $\operatorname{rank}_{\mathbb{F}_3}(C_{x_0})$ determine $\Delta$; they are $11, 15$ for $\mathrm{PG}(3, 2)$ and $15, 15$ for Bose. The finite verification is reproduced by the included script (Appendix D). □

# 6 Pasch switches and the selected quotient

A Pasch switch changes four triples on six points while preserving pair incidences. Its effect on the integral column lattice is controlled by the common saturation of the old and new four-block lattices.

**Theorem 6.1** (Pasch transport)**.** *Let $D, D'$ differ by a single Pasch (quadrilateral) switch. The two* local four-block lattices*—spanned by the four old, resp. four new, blocks that change—agree at every odd prime, and localized at 2 each has index two in their common saturation. Consequently every odd-primary part of $\Delta$ is preserved, the defect-order ratio $|\Delta(D')|/|\Delta(D)|$ lies in $\{1/2, 1, 2\}$ (so one of $\Delta(D), \Delta(D')$ may already equal the other), and the minimum numbers of binary generators of $\Delta(D)$ and $\Delta(D')$ differ by at most one.*

*Proof.* Four old blocks on six points are replaced by four new blocks. After labeling these points $0, \ldots, 5$, take the ordered old triples $012, 034, 135, 245$ and new triples $013, 024, 125, 345$. If $O, N$ are their $6 \times 4$ incidence matrices, then

$$N = \tfrac{1}{2}OH, \qquad H = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}, \qquad H^2 = 4I_4.$$

Both matrices have Smith factors $1, 1, 1, 2$, by elementary row and column reduction. Hence their column lattices $M_O, M_N$ have the same rational span, each has index two in its integral saturation $\widehat{M}$, and they agree over every $\mathbb{Z}_p$ with $p$ odd.

Since $2\widehat{M} \subseteq M_O \subseteq H_3$, also $\widehat{M} \subseteq H_3$.

Put $L_0$ for the lattice generated by all unaffected blocks. The exchange groups $E_O = \widehat{M}/M_O$ and $E_N = \widehat{M}/M_N$ are both $\mathbb{Z}/2$. The *global* defects are $\Delta(D) = H_k/(L_0 + M_O)$ and $\Delta(D') = H_k/(L_0 + M_N)$, formed only after adjoining the unchanged $L_0$; both surject onto the common quotient $H_k/(L_0 + \widehat{M})$ with kernel a quotient of $E_O$, resp. $E_N$. Hence every odd-primary part is preserved, the defect-order ratio lies in $\{1/2, 1, 2\}$, and the minimum binary generator numbers differ by at most one. Indeed, for $Q = H_3/(L_0 + \widehat{M})$ and either defect $F$, a kernel generated by at most one element gives $d_2(Q) \le d_2(F) \le d_2(Q) + 1$, where $d_2(F) = \dim_{\mathbb{F}_2}(F_2/2F_2)$. □

# 7 Geometric applications

The following families illustrate the quotient using classical incidence results. The first has zero defect; the second gives a homocyclic family.

## 7.1 Finite Möbius planes

A finite Möbius plane is a 3-$(q^2 + 1, q + 1, 1)$ design; its star differences are $q^2 - 1$ and $q$, with $\gcd = 1$, so $d = 1$ and Corollary 2.2 gives the full incidence Smith form $1^{\,q^2}(q+1)$ and $\Delta = 0$. This recovers the Miquelian and arbitrary even-order Smith forms of Ringe [28] from the star-difference bound.

## 7.2 Hermitian unitals

The classical Hermitian unital $H(q)$—the 2-$(q^3 + 1, q + 1, 1)$ design of Key's explicit finite-field/secant-line construction [26]—has, by Hiss's incidence divisors [27], the incidence cokernel

$$\operatorname{coker}(A(H(q))) \cong (\mathbb{Z}/(q+1))^{q^2-q+1}.$$

Removing the augmentation factor via Theorem 2.1 yields the homocyclic family

$$\Delta(H(q)) \cong (\mathbb{Z}/(q+1))^{q^2-q},$$

so $\Delta(H(4)) = (\mathbb{Z}/5)^{12}$. The incidence Smith form is Hiss's [27]; the displayed defect is its quotient by the augmentation factor.

A nonzero $t = 3$ instance—the design $\mathrm{SQS}(8) = \mathrm{AG}(3, 2)$, for which $d = 2$ and $\Delta \cong (\mathbb{Z}/2)^3$—is worked out in Appendix C, where it shows that the hypothesis $d = 1$ of Corollary 2.2 has content.

# A The STS(15) defect spectrum

This appendix records the complete realized defect spectrum on 15 points, the general squarefree trichotomy, and the ternary augmentation splitting for triple designs. The first two follow from the main results and classical $p$-rank theorems; the third is proved here and is used by the ternary case of Appendix B.

## A.1 The spectrum and primary purity

**Theorem A.1** (the STS(15) spectrum)**.** *For a Steiner triple system on* 15 *points,* $\operatorname{rank}_{\mathbb{F}_3}(A) = 14$ *and* $\operatorname{rank}_{\mathbb{F}_3}(C_{x_0}) = 15$*; hence* $\Delta = (\mathbb{Z}/2)^a$ *with* $a = 15 - \operatorname{rank}_{\mathbb{F}_2}(A)$*. The realized binary ranks are* $11, 12, 13, 14, 15$ *[25], so the complete realized spectrum is*

$$\Delta \in \{0,\ \mathbb{Z}/2,\ (\mathbb{Z}/2)^2,\ (\mathbb{Z}/2)^3,\ (\mathbb{Z}/2)^4\},$$

*with full incidence forms* $1^{14}3$, $1^{14}6$, $1^{13}2\cdot 6$, $1^{12}2^26$, $1^{11}2^36$ *respectively. More generally, using that a nontrivial* STS($v$) *has full $p$-rank for every prime $p \geq 5$ [15] together with the no-simultaneous-deficiency theorem of Shi–Xu–Krotov [16], every nontrivial* STS *defect is a pure* 2*-group, a pure* 3*-group, or zero.*

The STS(15) ternary statement is structural (a nonconstant ternary dual vector would partition the points into three STS(5)'s, impossible); the five binary ranks are imported from Assmus–Key [25] and Tonchev–Weishaar [18], and the code/affine-lattice classification of the 80 systems is Solé's [17]. The general trichotomy translates the cited $p$-rank results into the selected quotient.

### A.2 Removing the ternary augmentation factor

**Proposition A.2** (ternary augmentation splitting)**.** *For every nontrivial* 2-$(v, 3, \lambda)$ *design with $v > 3$, the all-ones block row is not in the $\mathbb{F}_3$-row span of the incidence matrix $A\colon \mathbb{Z}^{\mathcal{B}} \to \mathbb{Z}^P$; consequently the augmentation $\varepsilon\colon \operatorname{coker}(A) \to \mathbb{Z}/3$ splits, so $\operatorname{coker}(A) \cong \Delta(D) \oplus \mathbb{Z}/3$.*

*Proof.* Suppose $a \in \mathbb{F}_3^P$ satisfies $\sum_{x\in B} a_x = 1$ for every block $B$, and partition the points by fibre $P_i = \{x : a_x = i\}$, $n_i = |P_i|$ ($i \in \mathbb{Z}/3$). A triple sums to 1 in $\mathbb{F}_3$ only if its values form the multiset $(i, i, i+1)$; call such a block *type $i$*. Each pair inside $P_i$ lies in $\lambda$ blocks and forces the third point into $P_{i+1}$, giving $\lambda\binom{n_i}{2}$ blocks of type $i$; counting the same blocks by their two $P_i$–$P_{i+1}$ cross-pairs gives $2\cdot(\#\text{ type } i) = \lambda\, n_i n_{i+1}$. Hence $\lambda\, n_i(n_i-1) = \lambda\, n_i n_{i+1}$, so $n_{i+1} = n_i - 1$ whenever $n_i > 0$. Applied cyclically over $i \in \mathbb{Z}/3$ this forces three strict decreases $n_0 > n_1 > n_2 > n_0$, impossible unless some fibre is empty; and with an empty fibre the surviving instances of $n_{i+1} = n_i - 1$ bound $n_0 + n_1 + n_2 \leq 3$ (the maximum is the rotation $(0, 2, 1)$), contradicting $v > 3$.

Thus no such $a$ exists, so the all-ones row lies outside the $\mathbb{F}_3$-row span of $A$.

Row-space duality then supplies $y \in \mathbb{F}_3^{\mathcal{B}}$ with $Ay = 0$ in $\mathbb{F}_3^P$ and $\mathbf{1}_{\mathcal{B}}^T y = 1$. Lift $y$ to $\tilde{y} \in \mathbb{Z}^{\mathcal{B}}$; then $A\tilde{y} \equiv 0 \pmod 3$, so $z := A\tilde{y}/3 \in \mathbb{Z}^P$ is integral, its class $[z] \in \operatorname{coker}(A)$ satisfies $3[z] = [A\tilde{y}] = 0$, and, using $\mathbf{1}_P^T A = 3\,\mathbf{1}_{\mathcal{B}}^T$ (each block has three points),

$$\varepsilon([z]) = \mathbf{1}_P^T z \bmod 3 = \tfrac{1}{3}\mathbf{1}_P^T A\tilde{y} \bmod 3 = \mathbf{1}_{\mathcal{B}}^T \tilde{y} \bmod 3 = 1.$$

Hence $\mathbb{Z}/3 \to \operatorname{coker}(A)$, $1 \mapsto [z]$, is a section of $\varepsilon$, and $\operatorname{coker}(A) \cong \Delta(D) \oplus \mathbb{Z}/3$. □

## B Realization of elementary defects

The finite spectra above extend to infinite families. The key step is to compare exact-rank counting bounds while holding the desired defect fixed.

**Theorem B.1** (elementary realization)**.** *For every fixed $a \geq 1$, all sufficiently large $h$ admit an* STS($2^h - 1$) *with $\Delta \cong (\mathbb{Z}/2)^a$; for every fixed $c \geq 1$, all sufficiently large $h$ admit an* STS($3^h$) *with $\Delta \cong (\mathbb{Z}/3)^c$.*

*Proof.* We prove positivity of the exact-rank difference bounds of Jungnickel–Tonchev [19, Theorems 4.4 and 4.12] in the moving-parameter regime needed here. Their parameter $n$ is denoted by $h$ in this appendix, to distinguish it from $v - 1$.

**1. Counting inputs.** Let $N_1(q)$ count Steiner triple systems on a fixed $q$-point set, $N_2(q)$ count one-factorizations of the labeled $K_q$, and $N_3(q)$ count transversal designs on three specified groups of size $q$ (equivalently, labeled Latin squares). The required asymptotics are

$$\begin{aligned}
&\log N_1(q) = \tfrac{1}{6}q^2(\log q - 2 + o(1)) && (q \equiv 1, 3 \pmod 6),\\
&\log N_2(q) = \tfrac{1}{2}q^2(\log q - 2 + o(1)) && (q \text{ even}),\\
&\log N_3(q) = q^2(\log q - 2 + o(1)).
\end{aligned}$$

These are Keevash's enumeration theorem [21, Theorem 2.2], the complete-graph case of [22, Theorem 1.2] and its matching upper bound, and the permanent-bound Latin-square estimate recalled in the introduction of the latter paper. Ordering factors changes logarithms by at most $O(q \log q)$; exact formulas below use Jungnickel–Tonchev's conventions. Also $\log(q!) = O(q \log q)$.

**2. Ternary case.** Fix $c \geq 1$, put $t = h - c$, $T = 3^t$, and $M = 3^c$. For $h \geq c + 2$, the source condition $2 \leq t \leq h - 1$ holds. Set

$$F_3(T, M) = N_1(T)^M N_3(T)^{M(M-1)/6}.$$

The number of isomorphism classes of exact rank $3^h - h - 1 + t = v - 1 - c$ is at least $A_3 - B_3$, where

$$A_3 = \frac{F_3(T, M)}{(T!)^M |\operatorname{AGL}(c, 3)|}, \qquad B_3 = \frac{F_3(T/3, 3M)}{((T/3)!)^{3M-c-2}}.$$

All factorial exponents and group orders are fixed as $h \to \infty$. Consequently

$$\begin{aligned} \log A_3 &= \tfrac{1}{6} M^2 T^2 (\log T - 2 + o(1)), \\ \log B_3 &= \tfrac{1}{6} M^2 T^2 (\log T - \log 3 - 2 + o(1)), \\ \log(A_3/B_3) &= \tfrac{1}{6} M^2 T^2 \log 3 + o(T^2) > 0 \end{aligned}$$

for all sufficiently large $h$. This proves existence at the exact rank, not merely at a rank bounded above by it.

**3. Binary case.** Fix $a \geq 1$, put $t = h - a$, $S = 2^t$, $M = 2^a - 1$, and $Q = M + 1 = 2^a$. For $h \geq a + 2$ the same source condition holds. Define

$$F_2(S, M) = N_1(S - 1)\big(N_2(S)(S - 1)!\big)^M N_3(S)^{M(M-1)/6}.$$

The exact-rank-$2^h - h - 1 + t = v - a$ count is at least $A_2 - B_2$, with

$$A_2 = \frac{F_2(S, M)}{(S - 1)!(S!)^M |\operatorname{PGL}(a, 2)|}, \qquad B_2 = \frac{F_2(S/2, 2M + 1)}{((S/2)!)^{2M-a-1}}.$$

The coefficient of $S^2(\log S - 2)$ in $\log A_2$ is $1/6 + M/2 + M(M - 1)/6 = Q^2/6$. Replacing $S$ by $S - 1$ in $N_1$ and including the factorials changes this by only $o(S^2)$. For $B_2$ the group size is $S/2$ and the corresponding coefficient is $(2Q)^2/6$. Thus

$$\begin{aligned} \log A_2 &= \tfrac{1}{6} Q^2 S^2 (\log S - 2 + o(1)), \\ \log B_2 &= \tfrac{1}{6} Q^2 S^2 (\log S - \log 2 - 2 + o(1)), \\ \log(A_2/B_2) &= \tfrac{1}{6} Q^2 S^2 \log 2 + o(S^2) > 0 \end{aligned}$$

eventually, proving the required exact-rank existence.

**4. Conversion to the selected group.** For $v = 2^h - 1$, the annihilator $d = (v - 3)/2$ has $v_2(d) = 1$ for $h \geq 3$. Exact binary rank $v - a$ gives $a$ binary generators of $\Delta$ (the augmentation quotient has odd order). The primary purity in Appendix A excludes other primes, so $\Delta \cong (\mathbb{Z}/2)^a$. For $v = 3^h$, $v_3(d) = 1$ for $h \geq 2$. By Proposition A.2, $\operatorname{coker}(A) \cong \mathbb{Z}/3 \oplus \Delta$; thus rank $v - 1 - c$, whose incidence deficiency is $c + 1$, gives exactly $c$ ternary generators of $\Delta$. Purity and valuation one yield $\Delta \cong (\mathbb{Z}/3)^c$. □

The threshold in $h$ is not made effective here. The constructions and exact-rank bounds are imported; the displayed moving-parameter comparison and its translation to the selected quotient are the deductions supplied in this appendix. No non-elementary or mixed-primary realization is claimed.

## C A nonzero example: SQS(8)

The Steiner quadruple system $\mathrm{SQS}(8) = \mathrm{AG}(3,2)$ is a 3-$(8,4,1)$ design: its 14 blocks are the affine planes (cosets of the seven 2-dimensional subspaces) of $\mathbb{F}_2^3$. Its star sequence has $\lambda_1 = 7$, $\lambda_2 = 3$, $\lambda_3 = 1$, so the star differences are 4 and 2 and $d = \gcd\{4,2\} = 2 \neq 1$; Corollary 2.2 therefore does *not* force $\Delta = 0$. The pointed presentation is $8 \times 14$. Its Smith form is $1^5 2^3$, equivalently its $\binom{14}{8} = 3003$ maximal minors have gcd 8. Together with the exponent-two bound this gives

$$\Delta(\mathrm{SQS}(8)) \cong (\mathbb{Z}/2)^3, \qquad \text{full incidence Smith form } 1^4 2^3 4.$$

Here SQS(8) is the smallest nontrivial Steiner quadruple system, and this nonzero defect shows that the hypothesis $d = 1$ of Corollary 2.2 has genuine content: dropping it admits a nontrivial 2-group. The verifier reproduces the Smith forms and modular ranks, rather than enumerating these minors (Appendix D).

## D Reproducibility of the finite controls

The verifier `scripts/verify.py` constructs the following four designs from scratch, without external data. The middle column gives the full incidence Smith form; the last column gives the selected quotient.

| Design | Incidence Smith form | $\Delta(D)$ |
|---|---|---|
| $\mathrm{SQS}(8) = \mathrm{AG}(3,2)$ | $1^4 2^3 4$ | $(\mathbb{Z}/2)^3$ |
| Point–line $\mathrm{PG}(3,2)$ | $1^{11} 2^3 6$ | $(\mathbb{Z}/2)^4$ |
| Bose STS(15) | $1^{14} 3$ | 0 |
| Möbius plane $M(2)$ | $1^4 3$ | 0 |

Smith forms are computed with `python-flint`. A separate modular implementation checks the binary and ternary ranks of the pointed presentation. The output is checked against the frozen certificate `certificates/expected_verification.json` and is byte-identical under `python` and `python -O`.

The other cited finite data are imported: the larger Möbius examples [28], the projective-plane Smith forms [9], the Hermitian incidence divisors [27], and the intermediate STS(15) ranks [18, 16]. They are not recomputed by this verifier.

The finite computations support the two-system comparison in Theorem 5.2 and the worked example in Appendix C. The general arguments for the star bound, pointed presentation, local bilinear analysis, Pasch transport and elementary realization are written proofs, using the classical inputs cited in the text. Finite checks do not establish their uniform conclusions.

## Data and code availability

The companion repository contains the LaTeX source, compiled paper, verifier and frozen certificate. Appendix D describes the finite checks and their scope; `REPRODUCE.md` gives the commands and dependencies. The manuscript is licensed under CC BY 4.0; the original companion code is licensed under MIT.

## Use of artificial intelligence

The author used OpenAI Codex substantially in the mathematical exploration, development and checking of arguments, verification code, and manuscript drafting and revision. Exact model versions for the original research were not recorded. The October 2026 revision also used GPT-6.1

Sol review agents and a subsequent GPT-6 Astra review to examine the proofs, sources, and finite computations. The author directed the project and is responsible for the submitted claims. These AI-assisted checks and reproducible computations do not constitute independent specialist review or a formal verification of the general proofs.

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